AI and ML Algorithms that Power Self-Driving Cars

Self-driving cars are poised to become a reality in the very near future. These cars rely extensively on a range of AI and ML based algorithms to ply safely and reliably, and to make ethical decisions in the midst of traffic. Let's take a quick look at some of these algorithms.

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Self-driving cars exemplify the intersection of artificial intelligence (AI) and real-world applications. The technology behind self-driving cars relies on AI algorithms that enable vehicles to navigate roads, recognise objects, and make decisions based on the environment.

One of the most critical components of self-driving cars is computer vision, which uses AI algorithms to process and interpret visual data from cameras and sensors mounted on the vehicle. Computer vision enables self-driving cars to detect and classify objects on the road, including other vehicles, pedestrians, and traffic signs. This information is then used to determine the vehicle's speed, direction, and any necessary actions to avoid collisions.

Another essential component of self-driving cars is machine learning, a type of AI that allows the vehicle to learn from experience and improve its performance over time. Self-driving cars use machine learning algorithms to analyse vast amounts of data collected from sensors, cameras, and other sources. The algorithms then use this data to make predictions about the environment, such as the behaviour of other drivers or changes in road conditions.

Self-driving cars also depend on various other AI technologies, such as natural language processing, which allows the vehicle to understand spoken commands from passengers. Additionally, decision making algorithms empower the car to make informed decisions based on factors like road conditions, traffic patterns, and the vehicle's own capabilities.

Despite the progress made, there are significant challenges associated with the development and deployment of self-driving cars. Perhaps the most crucial is ensuring the safety and reliability of these vehicles. Self-